Database Backend for a WebGIS – City Festival

Final Report

Winter 2024/2025

1. **Introduction**

   Organizing a city festival requires a well-structured approach to managing various aspects, such as event locations, attractions, public facilities, and security arrangements. Given the dynamic nature of festivals, a traditional paper-based system or static database is insufficient. Instead, a spatial database backend integrated with a WebGIS application can provide real-time insights, enhance visitor experience, and improve operational efficiency.

   The festival_db database which is developed in PostgreSQL with PostGIS, serves as the backbone for managing the festival related spatial data. It supports location based queries, allowing users to find nearby attractions, upcoming events, and essential facilities such as first aid stations and restrooms. The system is designed with performance optimization, security, and scalability in mind, ensuring that it can handle large volumes of spatial data efficiently.

2. **Project Objectives**

   The primary goal of this project is to develop a fully functional spatial database that meets the following objectives:
   - **Efficient Data Management:** Design a normalized spatial database that organizes festival-related data, including events, facilities, attractions, and designated festival zones.
   - **Real-Time Spatial Querying:** Implement spatial queries that allow users to locate nearby facilities, attractions, and events dynamically based on their current location.
   - **WebGIS Integration:** Ensure that the database backend integrates smoothly with GIS tools like ArcGIS Pro and QGIS for real-time data visualization and analysis.
   - **Performance Optimization:** Utilize spatial indexing, query optimization, and database views to improve query efficiency and retrieval speed.
   - **Security and Access Control:** Implement security mechanisms to restrict access to sensitive data while allowing public users to retrieve non-sensitive festival information.

3. **Context and Use Case**

   Imagine working for Vudstork Entertainment Ltd, a company organizing a city-wide festival featuring concerts, amusement rides, food booths, and public facilities.

   The organizers need a centralized spatial database to:

- Store festival locations, schedules, and details for efficient management.
- Enable location-based queries so visitors can find nearby attractions and events.
- Identify essential facilities such as restrooms, security posts, and first aid stations.
- Support real-time GIS mapping and spatial analytics to improve crowd management and festival planning.

This database focuses exclusively on festival operations, excluding unrelated features like ticketing systems or financial data.

4. Data Modeling and Normalization

- **Entity Identification:** model core entities such as Festival_Info, Facility, Event, Attraction, and Zone.
- **Normalization:** All tables are designed in the third normal form (3NF) to eliminate redundancy and ensure consistency.
- **Spatial Data Types:** PostGIS geometry types (POINT and POLYGON) with SRID 4326 (WGS84) for all spatial attributes.

5. Database Design and Structure

The festival_db database is structured using relational modeling and spatial data management principles to ensure efficiency and maintainability.

5.1 Entity-Relationship Diagram (ERD)

The database structure consists of five key tables that store festival-related spatial data.

5.2 Table Descriptions and Relationships

Festival_Info Table

- Stores general details about the festival, such as name, start date, and description.
- Serves as the parent table, linking to events, facilities, and attractions.

Facility Table

- Stores spatial data for toilets, security posts, information booths, and charging stations.
- Uses POINT geometry to represent locations on a map.

Event Table

- Stores details about scheduled festival events with start and end times.
- Uses POINT geometry for event locations.

Attraction Table

- Stores details of festival attractions, such as food booths, amusement rides, and performance stages.
- Can use POINT or POLYGON geometry, depending on whether the attraction is a single location or an area.

Zone Table

- Defines festival zones, such as food areas, security zones, and concert spaces.
- Uses POLYGON geometry to enclose designated areas.

6. **Spatial Queries and Optimization**
   Several DML queries were used to test and analyze the efficiency of the database.

7. **Conclusion**
   By integrating advanced data modeling, spatial indexing, and real-time query capabilities, the solution not only meets the project requirements but also lays a solid foundation for scalable, user-friendly GIS applications.